情報数学 I, 第 4 回小テスト解答用紙　　2010/5/28 担当：那須

学籍番号 [ ][ ][ ][ ][ ][ ][ ]　氏名 解答例

・解答はこの用紙に記入．・答えには下線を引くなど，わかりやすくすること．

1 (1) (a) 奇関数　(b) 偶関数　(c) 奇関数　(d) 偶関数　(e) 偶関数 .

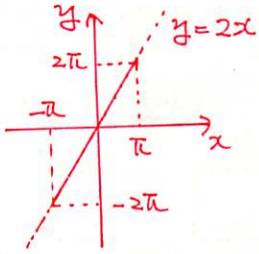
(2) ア 2　イ 0　(3) ウ 偶　エ 奇　オ 偶 .

2 (1) (a) $\int \cos\alpha x\, dx = \frac{1}{\alpha}\sin\alpha x + C$　(b) $\int \sin\beta x\, dx = -\frac{1}{\beta}\cos\beta x + C$.

(2) (c) $\sin n\pi = 0$　(d) $\cos n\pi = (-1)^n$　(3) (e) π　(f) 1　(g) L .

3 (1) $a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx$　$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx$.

(2)



$f(x)$は奇関数。従って $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \underline{\underline{0}}$ $(n=0,1,2\cdots)$

一方 $f(x)\sin nx$ は偶関数だから

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$$

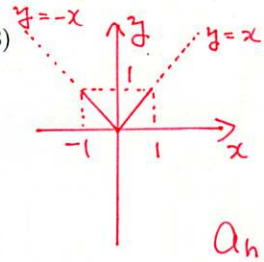
$$= \frac{2}{\pi}\int_0^{\pi} 2x\sin nx\,dx = \frac{4}{\pi}\int_0^{\pi} x\sin nx\,dx = \frac{4}{\pi}\left\{\left[x\left(-\frac{1}{n}\cos nx\right)\right]_0^{\pi}\right.$$

$$\left. - \int_0^{\pi}\left(-\frac{1}{n}\cos nx\right)dx\right\} = \frac{4}{\pi}\left\{-\frac{\pi}{n}\cos n\pi + \frac{1}{n}\int_0^{\pi}\cos nx\,dx\right\}$$

$$= \frac{4}{\pi}\left\{-\frac{\pi}{n}(-1)^n + \frac{1}{n}\left[\frac{1}{n}\sin nx\right]_0^{\pi}\right\} = -\frac{4}{n}(-1)^n = \underline{\underline{\frac{4}{n}(-1)^{n+1}}} \quad (n=1,2,\cdots)$$

ココまでで正解

(3)



$f(x) = |x|$は偶関数。

$$a_0 = \frac{1}{1}\int_{-1}^{1}|x|\,dx = 2\int_0^1 x\,dx = 2\left[\frac{x^2}{2}\right]_0^1 = \underline{\underline{1}}。$$ $n>0$のとき

$$a_n = \frac{1}{1}\int_{-1}^{1}|x|\cos n\pi x\,dx = 2\int_0^1 x\cos n\pi x\,dx$$

$$= 2\left\{\left[x\left(\frac{1}{n\pi}\sin n\pi x\right)\right]_0^1 - \int_0^1 \frac{1}{n\pi}\sin n\pi x\,dx\right\}$$

$$= -\frac{2}{n\pi}\int_0^1 \sin n\pi x\,dx = -\frac{2}{n\pi}\left[-\frac{1}{n\pi}\cos n\pi x\right]_0^1 = \underline{\underline{\frac{2}{n^2\pi^2}\left((-1)^n - 1\right)}}。$$

$(n=1,2,3,\cdots)$

$$b_n = \frac{1}{1}\int_{-1}^{1}\overbrace{|x|\sin nx}^{\text{奇関数}}\,dx = \underline{\underline{0}} \quad (n=1,2,3,\cdots)。$$

[0]※この講義に関する情報は FTP フォルダを参照．ftp://ftpi.sie.dendai.ac.jp/pub/nasu/Information Mathematics I/